In [84]:
```
import numpy as np
```

In [85]:
```
ones_arr = np.ones((5,5))
ones_arr
```

Out[85]:
```
array([[1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.]])
```

In [86]:
```
ones_arr = np.ones((5,5) ,dtype=int)
ones_arr
```

Out[86]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [87]:
```
zeros_arr = np.zeros((3,3), dtype=int)
zeros_arr
```

Out[87]:
```
array([[0, 0, 0],
       [0, 0, 0],
       [0, 0, 0]])
```

In [88]:
```
ones_arr
```

Out[88]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [89]:
```
ones_arr * 255
```

Out[89]:
```
array([[255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255]])
```

In [90]:
```
zeros_arr
```

Out[90]:
```
array([[0, 0, 0],
       [0, 0, 0],
       [0, 0, 0]])
```

In [91]:
```
ones_arr
```

Out[91]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [92]:
```
import matplotlib.pyplot as plt
```

In [93]:
```
%matplotlib inline
```

In [94]:
```
from PIL import Image   # pyhton imaging library
```

In [95]:
```
panda_img = Image.open("C:\Users\srini\Downloads\panda jpg.jpg")
```

```
  Cell In[95], line 1
    panda_img = Image.open("C:\Users\srini\Downloads\panda jpg.jpg")
                                                                   ^
SyntaxError: (unicode error) 'unicodeescape' codec can't decode bytes in position 2-
3: truncated \UXXXXXXXX escape
```

In [96]:
```
panda_img = Image.open(r"C:\Users\srini\Downloads\panda jpg.jpg")
```

In [97]:
```
panda_img
```

Out[97]:

In [98]:
```
type(panda_img)
```

Out[98]:
```
PIL.JpegImagePlugin.JpegImageFile
```

In [99]:
```
panda_arr=np.asarray(panda_img)
panda_arr
```

Out[99]:
```
array([[[130, 169,  52],
        [123, 161,  48],
        [117, 154,  51],
        ...,
        [ 42,  67,  25],
        [ 41,  69,   8],
        [ 95, 123,  48]],

       [[134, 173,  56],
        [125, 163,  50],
        [114, 151,  48],
        ...,
        [ 71,  97,  52],
        [ 79, 108,  44],
        [126, 154,  77]],

       [[135, 174,  55],
        [123, 161,  48],
        [108, 145,  42],
        ...,
        [ 93, 119,  71],
        [100, 129,  63],
        [127, 156,  76]],

       ...,

       [[115, 150,  34],
        [141, 175,  62],
        [147, 179,  72],
        ...,
        [ 98, 134,  12],
        [ 97, 131,  10],
        [105, 136,  16]],

       [[103, 138,  20],
        [122, 157,  41],
        [120, 152,  43],
        ...,
        [ 95, 130,  10],
        [ 88, 122,   2],
        [ 91, 122,   3]],

       [[ 96, 132,  10],
        [109, 144,  24],
        [106, 139,  26],
        ...,
        [ 93, 128,  10],
        [ 82, 115,   0],
        [ 81, 112,   0]]], shape=(148, 270, 3), dtype=uint8)
```

In [100…
```
type(panda_arr)
```

Out[100…
```
numpy.ndarray
```

In [101…
```
panda_arr.shape
```

Out[101…

```
(148, 270, 3)
```

In [102…

```
plt.imshow(panda_arr)
```

Out[102…

```
<matplotlib.image.AxesImage at 0x22474426350>
```

In [103…

```
panda_red=panda_arr.copy()
panda_red
```

Out[103…

```
array([[[130, 169,  52],
        [123, 161,  48],
        [117, 154,  51],
        ...,
        [ 42,  67,  25],
        [ 41,  69,   8],
        [ 95, 123,  48]],

       [[134, 173,  56],
        [125, 163,  50],
        [114, 151,  48],
        ...,
        [ 71,  97,  52],
        [ 79, 108,  44],
        [126, 154,  77]],

       [[135, 174,  55],
        [123, 161,  48],
        [108, 145,  42],
        ...,
        [ 93, 119,  71],
        [100, 129,  63],
        [127, 156,  76]],

       ...,

       [[115, 150,  34],
        [141, 175,  62],
        [147, 179,  72],
        ...,
        [ 98, 134,  12],
        [ 97, 131,  10],
        [105, 136,  16]],

       [[103, 138,  20],
        [122, 157,  41],
        [120, 152,  43],
        ...,
        [ 95, 130,  10],
        [ 88, 122,   2],
        [ 91, 122,   3]],

       [[ 96, 132,  10],
        [109, 144,  24],
        [106, 139,  26],
        ...,
        [ 93, 128,  10],
        [ 82, 115,   0],
        [ 81, 112,   0]]], shape=(148, 270, 3), dtype=uint8)
```

In [104…

```
panda_arr==panda_red
```

Out[104…

```
array([[[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       ...,

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]]], shape=(148, 270, 3))
```

In [105…

```
plt.imshow(panda_red)
```

Out[105…

```
<matplotlib.image.AxesImage at 0x2247448d090>
```

In [106…

```
panda_red.shape
```

Out[106…

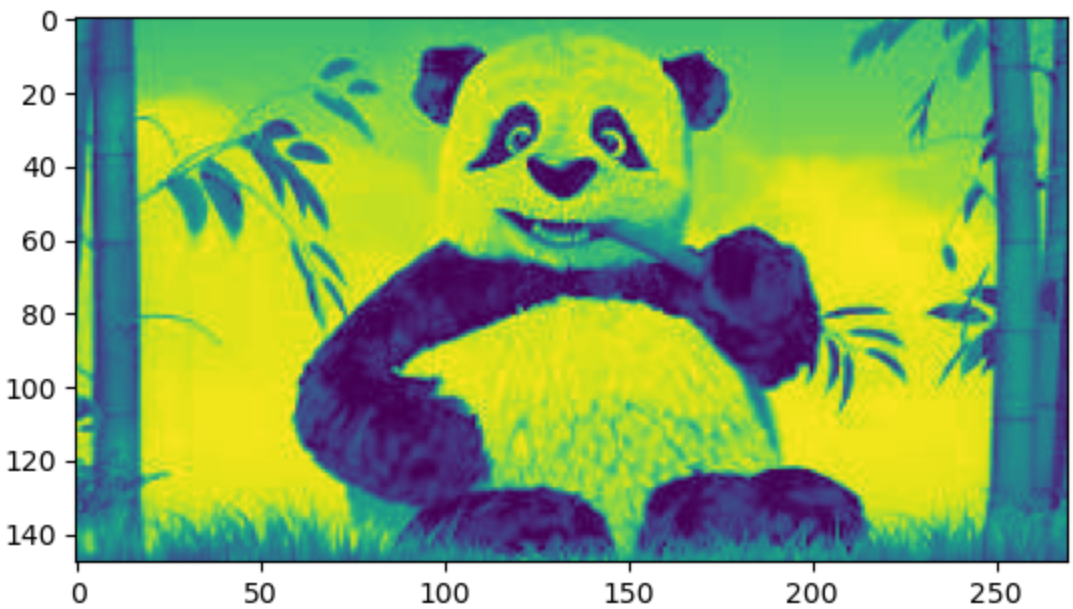
```
(148, 270, 3)
```

In [107…

```
# R G B
plt.imshow(panda_red[:,:,0])
```

Out[107…

```
<matplotlib.image.AxesImage at 0x224744dfd90>
```



In [108…

```
panda_red[:,:,0]
```

Out[108…

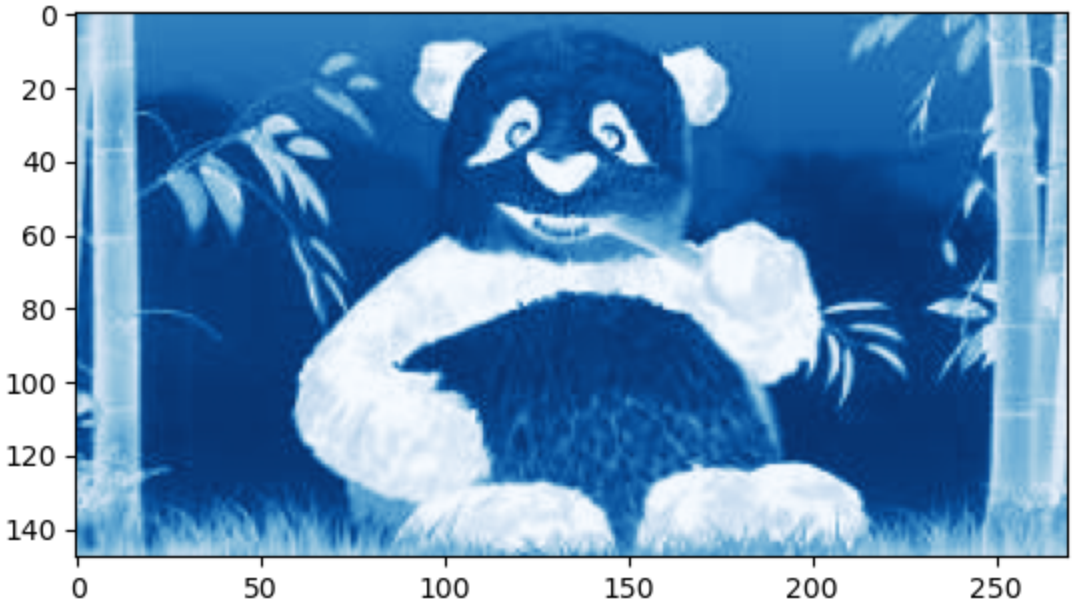
```
array([[130, 123, 117, ...,  42,  41,  95],
       [134, 125, 114, ...,  71,  79, 126],
       [135, 123, 108, ...,  93, 100, 127],
       ...,
       [115, 141, 147, ...,  98,  97, 105],
       [103, 122, 120, ...,  95,  88,  91],
       [ 96, 109, 106, ...,  93,  82,  81]], shape=(148, 270), dtype=uint8)
```

In [109…

```
plt.imshow(panda_red[:,:,0], cmap='Blues')
```

Out[109…

```
<matplotlib.image.AxesImage at 0x22474322ad0>
```

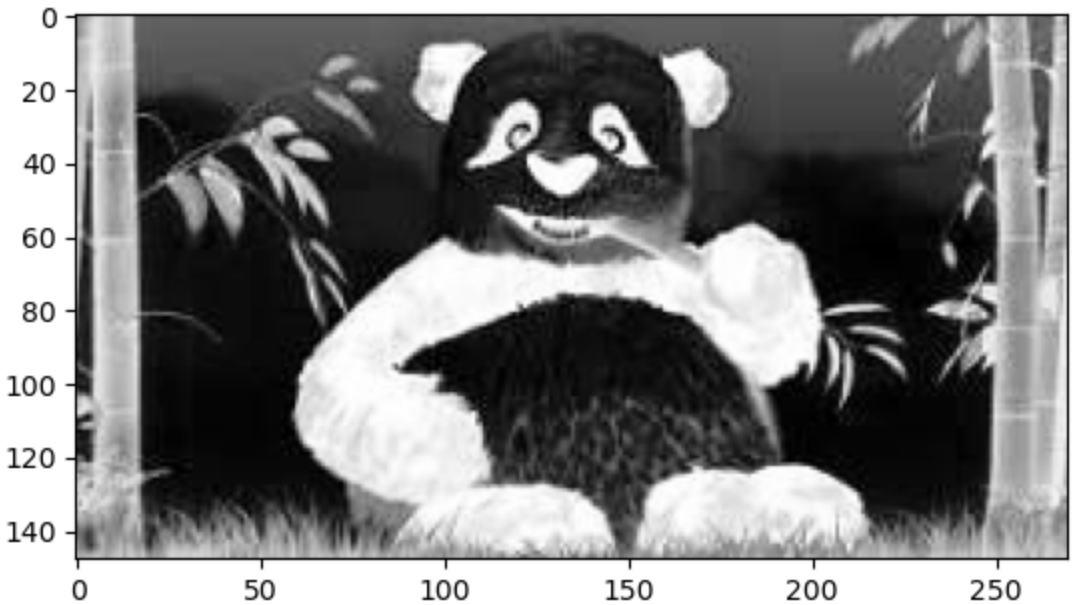


In [110…

```
plt.imshow(panda_red[:,:,0], cmap='Greys')
```

Out[110…

```
<matplotlib.image.AxesImage at 0x224743a1810>
```

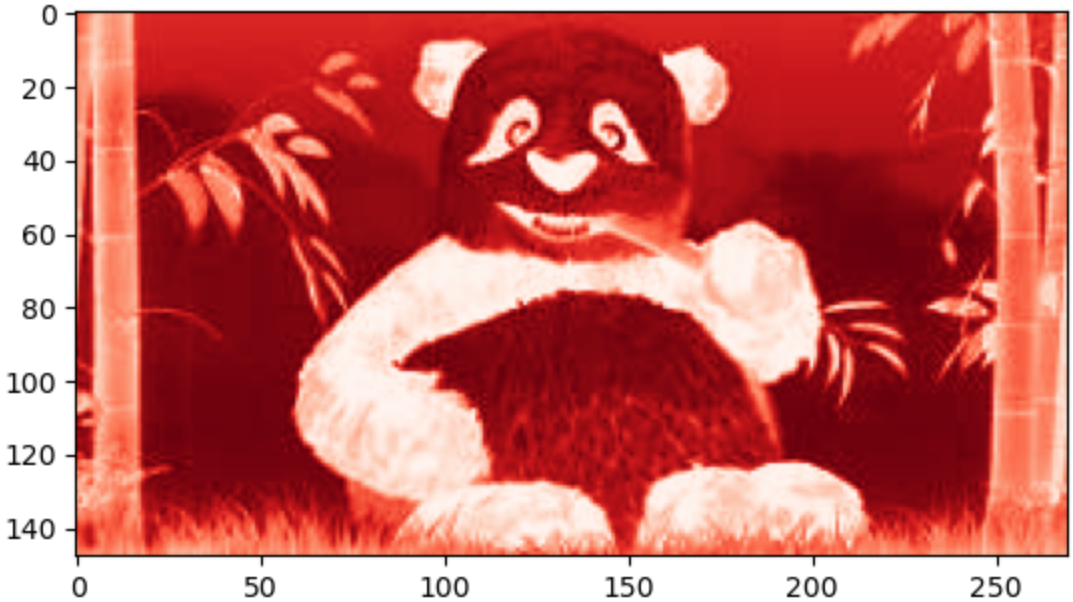


In [111…

```
plt.imshow(panda_red[:,:,0], cmap='Reds')
```

Out[111…

```
<matplotlib.image.AxesImage at 0x2247469c550>
```

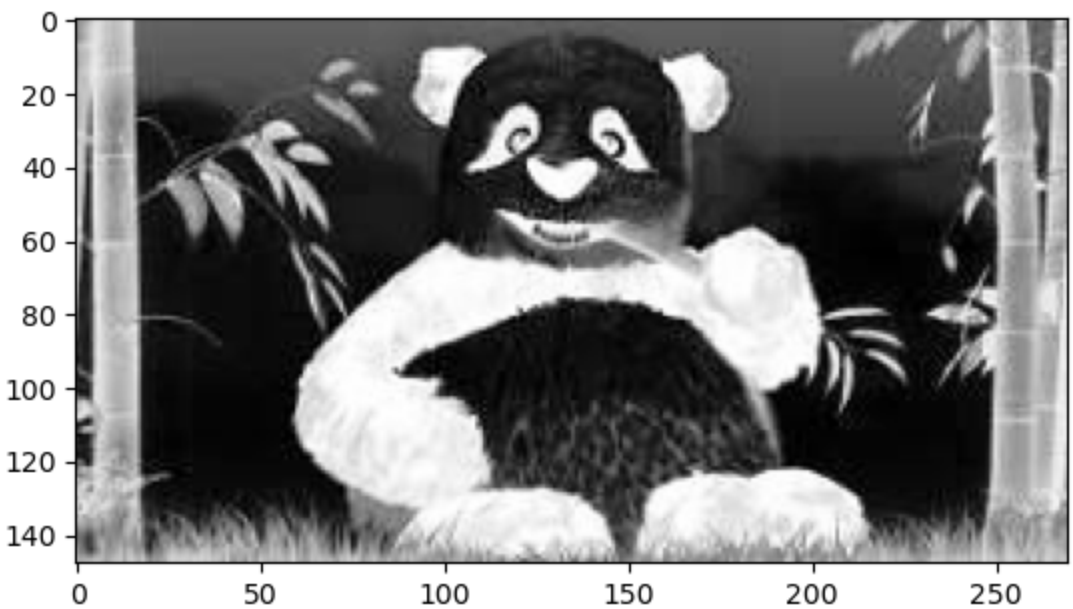


In [112…

```
plt.imshow(panda_red[:,:,0], cmap='Greys')
```

Out[112…

```
<matplotlib.image.AxesImage at 0x22474493250>
```

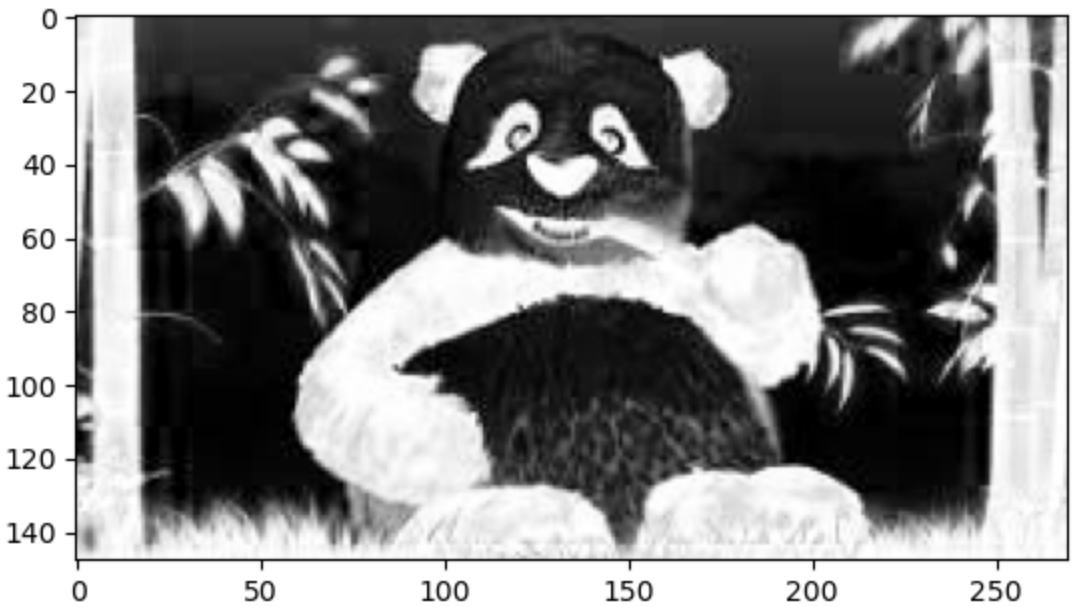


In [113…

```
plt.imshow(panda_red[:,:,2], cmap='Greys')
```

Out[113…

```
<matplotlib.image.AxesImage at 0x22474779f90>
```



In [114…

```
panda_red[:,:,0]
```

Out[114…

```
array([[130, 123, 117, ...,  42,  41,  95],
       [134, 125, 114, ...,  71,  79, 126],
       [135, 123, 108, ...,  93, 100, 127],
       ...,
       [115, 141, 147, ...,  98,  97, 105],
       [103, 122, 120, ...,  95,  88,  91],
       [ 96, 109, 106, ...,  93,  82,  81]], shape=(148, 270), dtype=uint8)
```

In [115…

```
panda_red[:,:,1]
```

Out[115…

```
array([[169, 161, 154, ...,  67,  69, 123],
       [173, 163, 151, ...,  97, 108, 154],
       [174, 161, 145, ..., 119, 129, 156],
       ...,
       [150, 175, 179, ..., 134, 131, 136],
       [138, 157, 152, ..., 130, 122, 122],
       [132, 144, 139, ..., 128, 115, 112]], shape=(148, 270), dtype=uint8)
```

In [116…

```
panda_red[:,:,2]
```

Out[116…

```
array([[52, 48, 51, ..., 25,  8, 48],
       [56, 50, 48, ..., 52, 44, 77],
       [55, 48, 42, ..., 71, 63, 76],
       ...,
       [34, 62, 72, ..., 12, 10, 16],
       [20, 41, 43, ..., 10,  2,  3],
       [10, 24, 26, ..., 10,  0,  0]], shape=(148, 270), dtype=uint8)
```

In [122…

```
panda_red[:,:,2]=0
```

In [123…

```
panda_red
```

Out[123…

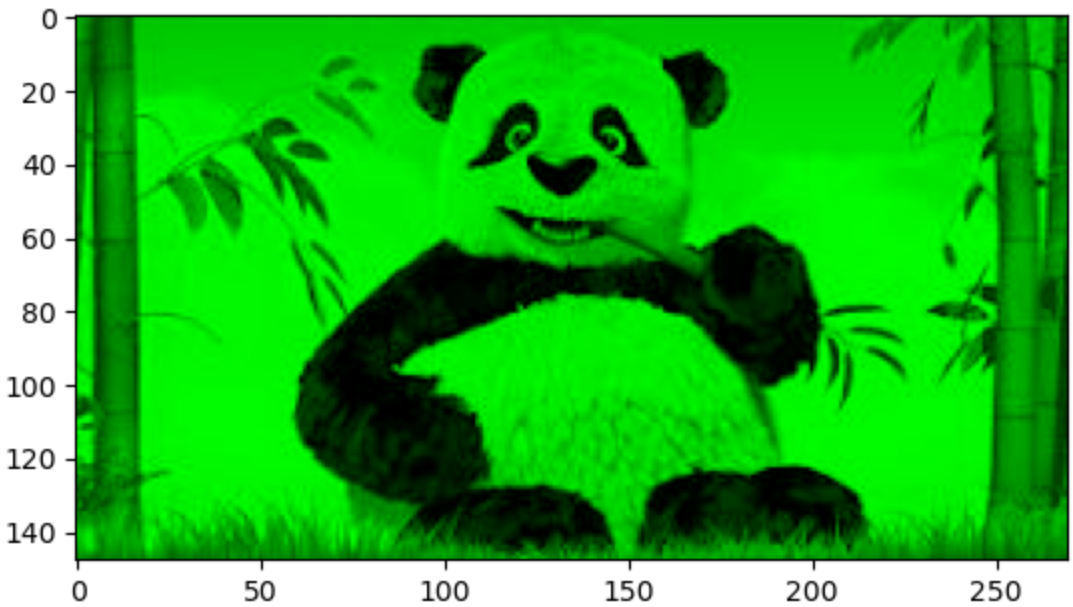
```
array([[[  0, 169,   0],
        [  0, 161,   0],
        [  0, 154,   0],
        ...,
        [  0,  67,   0],
        [  0,  69,   0],
        [  0, 123,   0]],

       [[  0, 173,   0],
        [  0, 163,   0],
        [  0, 151,   0],
        ...,
        [  0,  97,   0],
        [  0, 108,   0],
        [  0, 154,   0]],

       [[  0, 174,   0],
        [  0, 161,   0],
        [  0, 145,   0],
        ...,
        [  0, 119,   0],
        [  0, 129,   0],
        [  0, 156,   0]],

       ...,

       [[  0, 150,   0],
        [  0, 175,   0],
        [  0, 179,   0],
        ...,
        [  0, 134,   0],
        [  0, 131,   0],
        [  0, 136,   0]],

       [[  0, 138,   0],
        [  0, 157,   0],
        [  0, 152,   0],
        ...,
        [  0, 130,   0],
        [  0, 122,   0],
        [  0, 122,   0]],

       [[  0, 132,   0],
        [  0, 144,   0],
        [  0, 139,   0],
        ...,
        [  0, 128,   0],
        [  0, 115,   0],
        [  0, 112,   0]]], shape=(148, 270, 3), dtype=uint8)
```

In [124…

```
plt.imshow(panda_red)
```

Out[124…

```
<matplotlib.image.AxesImage at 0x22474686710>
```



In [141…
```
panda_red[:,:,0]
```

Out[141…
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], shape=(148, 270), dtype=uint8)
```

In [142…
```
panda_img
```

Out[142…

In [143…
```
arr1=np.asarray(panda_img)
```

In [144…
```
type(arr1)
```

Out[144…
```
numpy.ndarray
```

In [145…
```
arr1.shape
```

Out[145…
```
(148, 270, 3)
```

In [146…
```
plt.imshow(arr1)
```

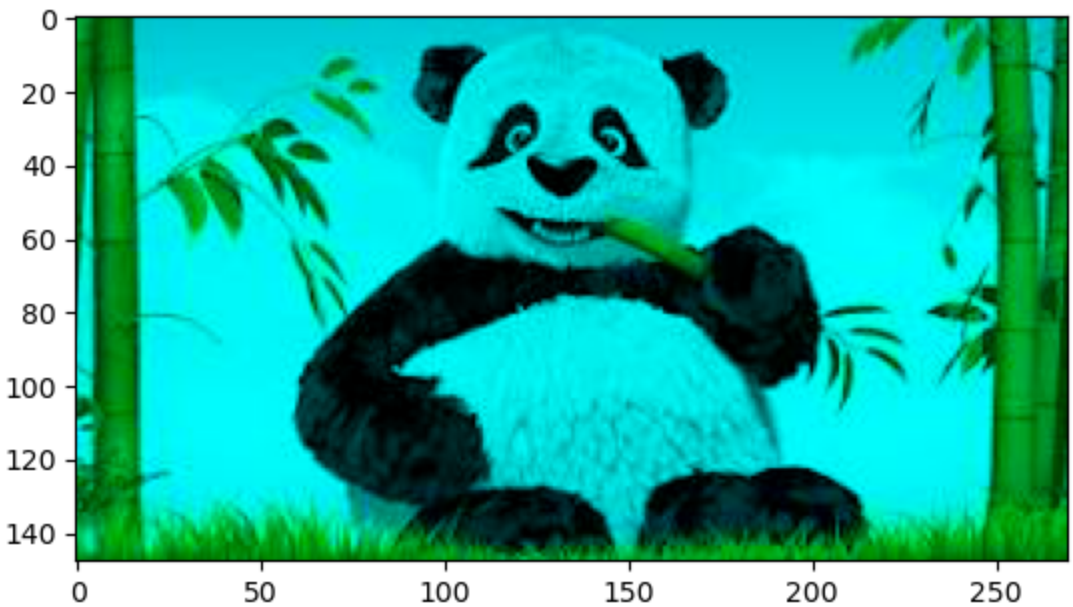
Out[146…
```
<matplotlib.image.AxesImage at 0x22474941450>
```

In [147…
```
panda_img1=arr1.copy()
```

In [149…
```
panda_img1[:,:,0]=0
```

In [150…
```
plt.imshow(panda_img1)
```

Out[150…
```
<matplotlib.image.AxesImage at 0x22474970190>
```



In [151…
```
panda_img1[:,:,1]
```

Out[151…

```
array([[169, 161, 154, ...,  67,  69, 123],
       [173, 163, 151, ...,  97, 108, 154],
       [174, 161, 145, ..., 119, 129, 156],
       ...,
       [150, 175, 179, ..., 134, 131, 136],
       [138, 157, 152, ..., 130, 122, 122],
       [132, 144, 139, ..., 128, 115, 112]], shape=(148, 270), dtype=uint8)
```

In [152…

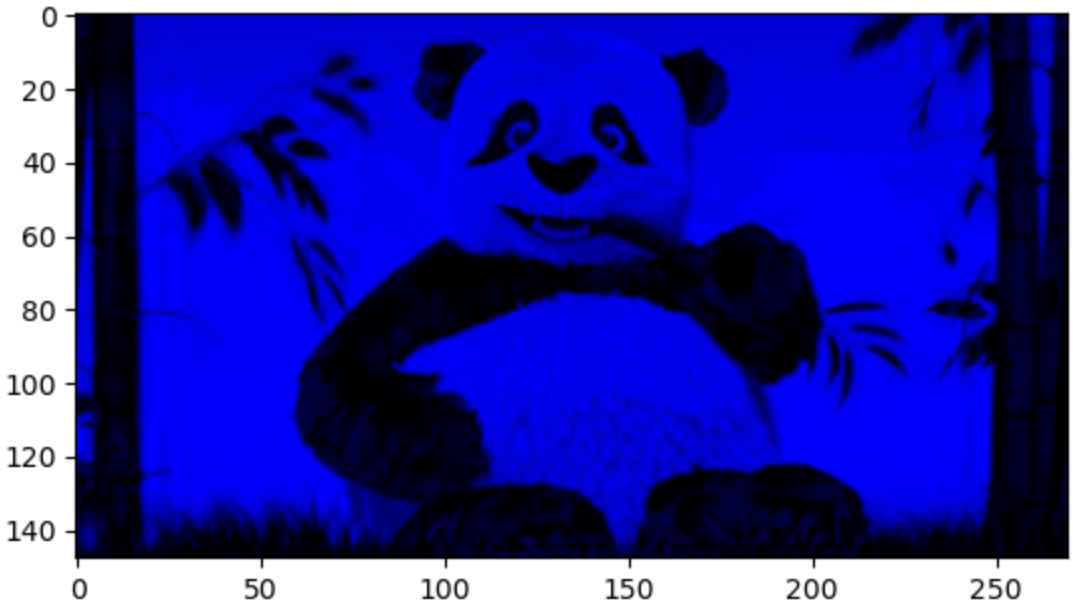
```
panda_img1[:,:,1]=0
```

In [153…

```
plt.imshow(panda_img1)
```

Out[153…

```
<matplotlib.image.AxesImage at 0x224749d6e90>
```



In [ ]: